

The $\alpha = 1$ Maz'ya endpoint: collapse to ordinary function-space inclusion

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Abstract

The source leaves the case $\alpha = 1$ in Maz'ya classes as one branch of a larger open program. We prove a sharp endpoint theorem for the inhomogeneous Sobolev-domain problem. A single bounded coiled exponential horn in $\mathcal{J}_1$ supports translated smooth bumps whose measures form a geometric scale. Consequently, for any Orlicz domain L^A , any rearrangement-invariant target Y , and every order $m \geq 1$, an embedding $W^m L^A(\Omega) \hookrightarrow Y(\Omega)$ valid for all $\Omega \in \mathcal{J}_1$ is equivalent to the ordinary inclusion $L^A(0, 1) \hookrightarrow Y(0, 1)$. Applied to the formal endpoint family $L^{\infty, q; -1-1/q}$, this gives nonexistence of a largest Orlicz domain for $q < \infty$, but the largest domain $\exp L$ when $q = \infty$.

1 Source question and precise scope

Beránek's paper treats measure-one Euclidean domains whose isoperimetric profiles satisfy $I_\Omega(t) \gtrsim t^\alpha$. Its reduction and optimality theorems assume $\alpha < 1$. The final remark explicitly leaves several much broader directions open:

Finally, by Theorem [3.5](#) there is no largest Orlicz space which would render the embedding [\(3.11\)](#) true for every Ω in $\mathcal{J}_\alpha$. $\square$

Remark 3.9. In this work, we have focused primarily on Maz'ya classes of Euclidean domains. We are aware that there are plenty of open questions worth pursuing which we leave open. Pivotal examples are the case $\alpha = 1$ in Maz'ya domains, Gaussian–Sobolev embeddings, embeddings on domains endowed with Frostman–Ahlfors measures, or embeddings on probability spaces. The reason we do not consider these cases is that the corresponding integral operators get too complicated (e.g. they take the form, at least for higher-order embeddings, of kernel-type operators). Hence such considerations reach beyond the scope of this text. We plan to study them in our following work.

The statement does not pose one theorem encompassing all four directions. This packet selects the precise first branch: universal, inhomogeneous Sobolev embeddings over the endpoint class $\mathcal{J}_1$. The result below also classifies the formal $\alpha = 1$ endpoint of the Lorentz–Zygmund target scale used in the source.

2 Definitions

All domains are bounded and have Lebesgue measure one. If $E \subset \Omega$ has finite perimeter, write $P(E, \Omega)$ for its relative perimeter. For $0 < t \leq \frac{1}{2}$, set

$$I_\Omega(t) = \inf\{P(E, \Omega) : E \subset \Omega, t \leq |E| \leq \tfrac{1}{2}\}.$$

The endpoint Maz'ya class is

$$\mathcal{J}_1 = \{\Omega : I_\Omega(t) \geq c_\Omega t \text{ for } 0 < t \leq \tfrac{1}{2}\}.$$

Let Y be a rearrangement-invariant (r.i.) Banach function space, with representation space $Y(0, 1)$. If A is a Young function, L^A denotes the corresponding Orlicz space with Luxemburg norm. We use the inhomogeneous norm

$$\|u\|_{W^m L^A(\Omega)} = \sum_{j=0}^m \|\nabla^j u\|_{L^A(\Omega)}.$$

The zeroth-order term is essential at this endpoint.

For $1 \leq q < \infty$, define

$$Y_q = L^{\infty, q; -1-1/q}(0, 1), \quad \|f\|_{Y_q}^q = \int_0^1 \frac{f^*(t)^q}{t [\log(2/t)]^{q+1}} dt,$$

with the usual equivalent r.i. norm. For $q = \infty$,

$$\|f\|_{Y_\infty} = \sup_{0 < t < 1} \frac{f^*(t)}{\log(2/t)}, \quad Y_\infty = \exp L.$$

3 Result and proof intuition

Theorem 1 (endpoint collapse). *Let $n \geq 2$, $m \geq 1$, let Y be an r.i. Banach function space on a probability space, and let A be a Young function. The following are equivalent.*

1. For every $\Omega \in \mathcal{J}_1$,

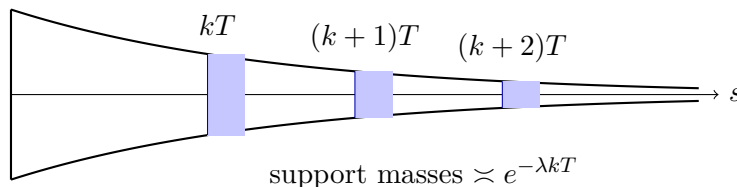
$$W^m L^A(\Omega) \hookrightarrow Y(\Omega),$$

where the embedding constant may depend on Ω .

2. $L^A(0, 1) \hookrightarrow Y(0, 1)$.

In fact, one fixed domain Ω_* suffices to deduce (2) from (1).

The mechanism is geometric. The domain Ω_* is a bounded tube around an infinite spiral, with exponentially shrinking cross-section. Moving a fixed smooth bump one unit farther along the tube multiplies the measure of its support by a fixed factor, without changing the size of any derivative. The coiling keeps the domain bounded while preserving uniform longitudinal geometry. Disjoint translated bumps therefore reproduce every geometrically discretized measurable function. A universal Sobolev embedding on $\mathcal{J}_1$ must already bound all such functions before differentiation is used.



4 The bounded coiled exponential horn

We first build a bounded curve of infinite length with uniform local geometry. For large t , consider the planar spiral

$$\tilde{\gamma}(t) = ((1 + (t + 2)^{-1}) \cos t, (1 + (t + 2)^{-1}) \sin t, 0, \dots, 0).$$

Reparametrize a tail of this curve by arclength, writing it as $\gamma : [0, \infty) \rightarrow \mathbb{R}^n$. The curve stays in a bounded annulus, has infinite length, and all its arclength derivatives are bounded. Successive turns at parameter size s are separated by a distance comparable to s^{-2} . Choose a smooth orthonormal normal frame $N_1(s), \dots, N_{n-1}(s)$ with bounded derivatives and put

$$\varepsilon(s) = \varepsilon_0 e^{-\beta s}, \quad F(y, s) = \gamma(s) + \varepsilon(s) \sum_{i=1}^{n-1} y_i N_i(s), \quad y \in B, s > 0,$$

where B is the unit ball in $\mathbb{R}^{n-1}$. For sufficiently small $\varepsilon_0 > 0$, exponential decay is faster than the polynomial separation of the turns, so F is injective. Its image Ω_0 is a bounded open connected tube. After one fixed homothety, call the resulting measure-one domain Ω_* . This homothety affects only fixed constants below.

Lemma 2. Ω_* is bounded, $|\Omega_*| = 1$, and $I_{\Omega_*}(t) \asymp t$ for $0 < t \leq \frac{1}{2}$. In particular, $\Omega_* \in \mathcal{J}_1$.

Proof. Let $\lambda = \beta(n - 1)$. The standard tube-Jacobian formula and bounded curvature give, uniformly in (y, s) ,

$$J_F(y, s) = \varepsilon(s)^{n-1} (1 + O(\varepsilon(s))) \asymp e^{-\lambda s}.$$

Thus $0 < |\Omega_0| < \infty$, and the announced homothety normalizes its volume.

The uniform probability measure on the bounded ball B satisfies an L^1 Poincaré inequality, as does the one-sided exponential probability measure $e^{-\lambda s} ds$. The elementary Fubini argument for the mean in the two variables tensorizes these inequalities. Since J_F is comparable to the product density and the differential of F is uniformly bounded, the same argument applied to $g = v \circ F$ gives

$$\int_{\Omega_*} |v - v_{\Omega_*}| dx \leq C \int_{\Omega_*} |\nabla v| dx.$$

Approximation of χ_E by smooth functions yields

$$2|E|(1 - |E|) \leq CP(E, \Omega_*).$$

Thus $P(E, \Omega_*) \gtrsim |E|$ when $|E| \leq \frac{1}{2}$, and $I_{\Omega_*}(t) \gtrsim t$.

For the reverse estimate, take the coiled tail $E_R = F(B \times (R, \infty))$. Its volume and the relative perimeter of its cross-sectional cut are both comparable to $e^{-\lambda R}$. Since R ranges continuously, $I_{\Omega_*}(t) \lesssim t$ for every $0 < t \leq \frac{1}{2}$. $\square$

5 Geometric bump transference

Lemma 3. Fix $m \geq 1$. If $W^m L^A(\Omega_*) \hookrightarrow Y(\Omega_*)$, then $L^A(0, 1) \hookrightarrow Y(0, 1)$.

Proof. Choose $0 < \ell < T$ and a nonnegative $\theta \in C_c^\infty(0, \ell)$ which is one on a nonempty inner interval. Put $\rho = e^{-\lambda T}$. For a finite nonnegative sequence (a_k) , supported at large positive integers, define

$$u(F(y, s)) = \sum_k a_k \theta(s - kT).$$

The summands have disjoint supports. With a Bishop normal frame, the longitudinal coordinate satisfies

$$\nabla s = \frac{\gamma'(s)}{1 - \varepsilon(s)\kappa(s) \cdot y}.$$

The spiral and all its arclength derivatives are bounded, while $\varepsilon'/\varepsilon = -\beta$. The standard tubular-coordinate formulas therefore show that every fixed higher derivative of s is uniformly bounded on the tube. Consequently, each support has measure $c_0\rho^k$, each plateau on which the corresponding summand equals a_k has measure $c_1\rho^k$, and for $0 \leq j \leq m$,

$$|\nabla^j u| \leq C_j \sum_k a_k \chi_{S_k}.$$

Let $I_k = (\rho^{k+1}, \rho^k]$ and $g_a = \sum_k a_k \chi_{I_k}$. Multiplying all atom measures by a fixed constant is a fixed dilation of the decreasing rearrangement. Dilation operators are bounded on every r.i. space. The lattice property, the plateau lower bound, and the derivative upper bounds therefore give constants independent of (a_k) such that

$$\|g_a\|_Y \lesssim \|u\|_{Y(\Omega_*)}, \quad \|u\|_{W^m L^A(\Omega_*)} \lesssim \|g_a\|_{L^A}.$$

The assumed Sobolev embedding implies

$$\|g_a\|_Y \lesssim \|g_a\|_{L^A} \tag{1}$$

for every finite geometric step function.

For completeness, these steps determine the full inclusion. If $f = f^*$ is nonnegative and supported near zero, let g equal $f(\rho^{k+1})$ on I_k . Then

$$f(t) \leq g(t) \leq f(\rho t).$$

The right-hand function is a fixed dilation of f . Truncating g , applying (1), and using the Fatou and dilation properties yields $\|f\|_Y \lesssim \|f\|_{L^A}$. The part away from zero is bounded and is controlled in every Banach function space. Rearrangement invariance now gives $L^A(0, 1) \hookrightarrow Y(0, 1)$. $\square$

Proof of Theorem 1. Condition (1) applied to the fixed horn $\Omega_* \in \mathcal{J}_1$ implies condition (2) by Lemma 3. Conversely, if $L^A(0, 1) \hookrightarrow Y(0, 1)$, then for every measure-one domain

$$\|u\|_{Y(\Omega)} \lesssim \|u\|_{L^A(\Omega)} \leq \|u\|_{W^m L^A(\Omega)}.$$

This proves the equivalence. $\square$

6 Optimal Orlicz domains at the endpoint

The principal alternative for Orlicz subspaces says that, for an r.i. space Y , let $L(Y)$ be the unique Orlicz space with the same fundamental function. There is a largest Orlicz space contained in Y exactly when $L(Y) \hookrightarrow Y$; in that case it is $L(Y)$. Combined with Theorem 1, this immediately gives the endpoint Sobolev classification.

Corollary 4. *Let $m \geq 1$ and $1 \leq q \leq \infty$.*

1. *If $q < \infty$, there is no largest Orlicz space L^A such that*

$$W^m L^A(\Omega) \hookrightarrow L^{\infty, q; -1-1/q}(\Omega) \quad \text{for every } \Omega \in \mathcal{J}_1.$$

2. If $q = \infty$, the largest such Orlicz space is $\exp L$.

Proof. For every q , the fundamental function of Y_q is

$$\varphi_{Y_q}(t) \asymp \frac{1}{\log(2/t)},$$

so $L(Y_q) = \exp L$. If $q < \infty$, the decreasing function $h(t) = \log(2/t)$ belongs to $\exp L$, since $\sup h(t)/\log(2/t) = 1$, but

$$\|h\|_{Y_q}^q = \int_0^1 \frac{dt}{t \log(2/t)} = \infty.$$

Hence $\exp L \not\hookrightarrow Y_q$, and the principal alternative gives nonexistence of a largest Orlicz subspace, hence of a largest endpoint Sobolev domain. When $q = \infty$, the defining norm gives $Y_\infty = \exp L$, so it is itself the largest domain. $\square$

7 Verification, novelty, and limitations

The proof has four independent checks. First, the bounded coiled horn has normalized volume one and tail volume/perimeter comparable to $e^{-\lambda R}$. Second, its lower isoperimetric bound is forced by a transported tensor-product L^1 Poincaré inequality. Third, all derivatives of the translated bump family live on the same geometric atoms, so the argument is uniform in the order m . Fourth, the finite- q endpoint witness reduces exactly to the divergent logarithmic integral above.

Targeted searches of the source, its published version, the cited general Orlicz-optimality papers, and phrase/keyword indexes found no statement of the horn-collapse equivalence or Corollary 4. Literature novelty is nevertheless the main point requiring expert review.

This is a promoted partial result because the source's final remark is a multi-branch program. The theorem concerns inhomogeneous $W^m L^A$ spaces; without the zeroth-order term, homogeneous $V^m L^A$ spaces require a different analysis. It classifies embeddings required on every domain in $\mathcal{J}_1$, not the sharper behavior of one fixed regular domain. It does not settle the Gaussian, Frostman–Ahlfors, or general probability-space directions.

Human review recommendation. Verify the tensorized L^1 Poincaré-to-isoperimetry passage on the horn, the geometric-step discretization in Lemma 3, and literature novelty. The finite- q /infinite- q classification and all endpoint exponents should then be checked against the published notation for Lorentz–Zygmund spaces.

References

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